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Department of Mathematics
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Math 101 Major Exam II Term 251

Original exam date: 11 November 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If all edges of a cube are expanding at a rate of 6 cm/sec, then the surface area is changing, when each edge is 2 cm, at the rate of

(a) 24 cm^2/sec

(b) 36 cm^2/sec

(c) 72 cm^2/sec

(d) 144 cm^2/sec

(e) 96 cm^2/sec

2. The graph of the equation $4x^2 + y^2 - 8x + 4y + 4 = 0$ has a horizontal tangent line at the point

(a) $(1, 1)$

(b) $(-2, 0)$

(c) $(2, -2)$

(d) $(0, -2)$

(e) $(1, -4)$

3. If $f(x) = \sqrt[3]{x}\,(\sqrt{x} + 3)$, then $f'(x) =$

(a) $\dfrac{5x^{2/3} + 6x^{1/6}}{6x^{5/6}}$

(b) $\dfrac{5x^{2/3} + x^{1/6}}{6x^{5/6}}$

(c) $\dfrac{5x^{1/6} + 6x^{2/3}}{6x^{5/6}}$

(d) $\dfrac{x^{2/3} + x^{1/6}}{6x^{5/6}}$

(e) $\dfrac{x^{2/3} + 6x^{2/3}}{6x^{5/6}}$

4. Using Newton's method to approximate the zero of the function $f(x) = x - e^{-x}$ with initial value $x_1 = 0$, $x_2 =$

(a) 0

(b) $\frac{1}{2}$

(c) $-\frac{1}{4}$

(d) 1

(e) $-\frac{1}{2}$

5. The equation of the tangent line to the graph of $x \arctan y + x^2 = y - 1$ at $(0, 1)$ is

(a) $4y + \pi x = 2$

(b) $y - 4\pi x = 4$

(c) $y + \frac{\pi}{4}x = 1$

(d) $y = \frac{\pi}{4}x + 4$

(e) $4y - \pi x = 4$

6. If an airplane flies at an altitude of 5 kms toward a point directly over an observer (see figure below) and the speed of the plane is 400 kms per hour, then the rate at which the angle of elevation θ is changing when the angle is 30° is equal to

(a) 5 rad/hr

(b) 80 rad/hr

(c) 20 rad/hr

(d) 40 rad/hr

(e) 10 rad/hr

7. The slope of the graph of $(\sin \pi x + \cos \pi y)^2 = 1$ at the point $\left(\frac{1}{6}, \frac{1}{3}\right)$ is equal to

(a) The graph has no slope at this point

(b) 1

(c) $\frac{1}{2}$

(d) $\frac{\sqrt{3}}{2}$

(e) 0

8. Let M and m be the absolute Maximum and absolute Minimum, respectively, for the function $f(x) = e^x \cos x$ on the interval $[0, \pi]$. Then $\dfrac{M^4}{m} =$

(a) $\dfrac{-e^{\pi/4}}{4}$

(b) $\dfrac{-e^{\pi/2}}{4}$

(c) $\dfrac{-1}{4}$

(d) $\dfrac{-1}{2}$

(e) $\dfrac{-e^{\pi}}{4}$

9. If $f(x) = x - \dfrac{3}{2}x^{2/3}$, then the number of critical numbers is

(a) 4

(b) 3

(c) 2

(d) 1

(e) 0

10. If $y = \cos\sqrt{\sin(\tan \pi x)}$, then $y'\left(\dfrac{1}{4}\right) =$

(a) $\dfrac{-\pi \csc\left(\sqrt{\sin 1/2}\right)\cos 1}{\sqrt{\sin 1}}$

(b) $\dfrac{-\pi \sin\left(\sqrt{\sin 1}\right)\sin 1}{\sqrt{\cos 1}}$

(c) $\dfrac{-\pi \sin\left(\sqrt{\sin 1}\right)\cos 1}{\cos\sqrt{\sin 1}}$

(d) $\dfrac{-\pi \sin\left(\sqrt{\sin 2}\right)\cos 1}{\sqrt{2}\cos\sqrt{\sin 1}}$

(e) $\dfrac{-\pi \sin\left(\sqrt{\sin 1}\right)\cos 1}{\sqrt{\sin 1}}$

11. If $y = \arctan x - \dfrac{x}{1+x^2}$, then $y'(x) =$

(a) $\dfrac{2x}{(1+x^2)^2}$

(b) $\dfrac{1-x^2}{1+x^2}$

(c) $\dfrac{1-x^2}{(1+x^2)^2}$

(d) $\dfrac{2}{(1+x^2)^2}$

(e) $\dfrac{2x^2}{(1+x^2)^2}$

12. The value of c that satisfies Rolle's Theorem when applied to $f(x) = (x-4)(x+2)^2$ on $[-2, 4]$ is equal to

(a) -2

(b) 1

(c) 2

(d) 0

(e) -1

13. The value of c that satisfies the Mean Value Theorem when applied to $f(x) = \dfrac{x}{x+1}$ on $\left[-\dfrac{1}{2}, 2\right]$ is equal to

(a) $-1 - \dfrac{\sqrt{6}}{2}$

(b) $-1 + \dfrac{\sqrt{6}}{2}$

(c) $-1 - \dfrac{\sqrt{3}}{2}$

(d) $-1 + \dfrac{\sqrt{3}}{2}$

(e) $\dfrac{2}{3}$

14. If $y - \cos(xy) = x$, then $y''(0) =$

(a) 2π

(b) -2

(c) 2

(d) -1

(e) 0

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$S = 6x^2$, $S' = 12x\,x' = 12(2)(6) = 144$ cm^2/sec.
2	**(e)**	$y' = \frac{4(1-x)}{y+2} = 0 \Rightarrow x = 1$; then $y^2 + 4y = 0$, and $(1, -4)$ is the listed point.
3	**(a)**	$f = x^{5/6} + 3x^{1/3}$, so $f' = \frac{5}{6}x^{-1/6} + x^{-2/3} = \frac{5x^{2/3}+6x^{1/6}}{6x^{5/6}}$.
4	**(b)**	$f' = 1 + e^{-x}$; $x_2 = 0 - \frac{f(0)}{f'(0)} = 0 - \frac{-1}{2} = \frac{1}{2}$.
5	**(e)**	$y'(0,1) = \arctan 1 = \frac{\pi}{4}$; line $y - 1 = \frac{\pi}{4}x$, i.e. $4y - \pi x = 4$.
6	**(c)**	$\tan\theta = 5/x$, $x' = -400$: $\frac{4}{3}\theta' = \frac{2000}{75}$, so $\theta' = 20$ rad/hr.
7	**(b)**	$2(\sin\pi x + \cos\pi y)\pi(\cos\pi x - \sin\pi y\, y') = 0$; at the point $\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}y' = 0$, $y' = 1$.
8	**(c)**	Critical point $\frac{\pi}{4}$: $M = \frac{e^{\pi/4}}{\sqrt{2}}$, $m = -e^{\pi}$, so $\frac{M^4}{m} = -\frac{1}{4}$.
9	**(c)**	$f' = \frac{x^{1/3}-1}{x^{1/3}}$: zero at $x = 1$, undefined at $x = 0$ (in domain), so 2 critical numbers.
10	**(e)**	Chain rule; at $x = \frac{1}{4}$, tan $= 1$, sec$^2 = 2$: $y' = \frac{-\pi\sin(\sqrt{\sin 1})\cos 1}{\sqrt{\sin 1}}$.
11	**(e)**	$y' = \frac{1}{1+x^2} - \frac{1-x^2}{(1+x^2)^2} = \frac{2x^2}{(1+x^2)^2}$.
12	**(c)**	$f' = 3(x+2)(x-2)$; $c = \pm 2$, and only $c = 2$ is in $(-2, 4)$.
13	**(b)**	Average slope $\frac{2}{3}$; $\frac{1}{(c+1)^2} = \frac{2}{3}$, $c = -1 + \frac{\sqrt{6}}{2}$ (the other root is outside).
14	**(d)**	$y(0) = 1$, $y'(0) = 1$; second derivative gives $y'' + \cos(xy)(y + xy')^2 + \cdots = 0$, so $y''(0) = -1$.